

Scene Setup Guide

This document explains how to configure the three Unity scenes after opening the project in the Unity Editor. The game is designed to self-bootstrap at runtime via Setup scripts, but the scenes themselves need a minimal skeleton.

Scene 1: MainMenu (Assets/Scenes/MainMenu.unity)

1. **Create the scene:** File > New Scene > Basic (Built-in) .
2. **Save as:** Assets/Scenes/MainMenu.unity .
3. **Create an empty GameObject** named MainMenuBootstrap .
4. **Attach** the MainMenuSetup script to it.
5. The MainMenuSetup.Awake() method creates the camera config, background, canvas, title, buttons, and wires the MainMenuUI component automatically.
6. **Delete** the default Directional Light (2D game — not needed).

Objects in scene:

GameObject	Components
Main Camera	Camera (auto-configured)
MainMenuBootstrap	MainMenuSetup

Scene 2: Game (Assets/Scenes/Game.unity)

1. **Create the scene:** File > New Scene > Basic (Built-in) .
2. **Save as:** Assets/Scenes/Game.unity .
3. **Create an empty GameObject** named GameBootstrap .
4. **Attach** the GameSetup script to it.
5. GameSetup.Awake() creates:
 - Player (with sprites, physics, health, shooting, shield visual)
 - Enemy prefab templates (4 types)
 - Bullet prefab templates (player + enemy)
 - Power-up prefab templates (3 types)
 - EnemySpawner
 - 3-layer procedural parallax background
 - Full HUD canvas (health bar, shield bar, score, combo, wave, weapon level)
 - Pause menu overlay
6. **Delete** the default Directional Light.

Objects in scene:

GameObject	Components
Main Camera	Camera (auto-configured)
GameBootstrap	GameSetup

Scene 3: GameOver (Assets/Scenes/GameOver.unity)

1. **Create the scene:** File > New Scene > Basic (Built-in) .
2. **Save as:** Assets/Scenes/GameOver.unity .
3. **Create an empty GameObject** named GameOverBootstrap .
4. **Attach** the GameOverSetup script to it.
5. GameOverSetup.Awake() creates the game-over canvas, score displays, new-high-score indicator, and Retry / Main Menu buttons.
6. **Delete** the default Directional Light.

Objects in scene:

GameObject	Components
Main Camera	Camera (auto-configured)
GameOverBootstrap	GameOverSetup

Build Settings

After creating all three scenes, go to File > Build Settings and add them in this order:

Index	Scene
0	Assets/Scenes/MainMenu.unity
1	Assets/Scenes/Game.unity
2	Assets/Scenes/GameOver.unity

Tag & Layer Setup

The project's TagManager.asset already defines the required tags and layers. If they don't appear automatically, create them manually:

Tags:

- Player
- PlayerBullet
- EnemyBullet
- PowerUp
- Enemy
- Background
- Boundary

Layers (user layers starting at index 8):

- Layer 8: Player
- Layer 9: Enemy
- Layer 10: PlayerBullet
- Layer 11: EnemyBullet
- Layer 12: PowerUp
- Layer 13: Background
- Layer 14: Boundary

Sorting Layers:

- Background
- Midground
- Foreground
- Player
- Effects
- UI

Audio Setup (Optional)

Place audio files in `Assets/Resources/Audio/` :

```
Assets/  
  Resources/  
    Audio/  
      Music/  
        menu_music.wav      (or .mp3, .ogg)  
        game_music.wav  
      SFX/  
        player_shoot.wav  
        enemy_shoot.wav  
        explosion.wav  
        powerup.wav  
        hit.wav  
        shield_up.wav  
        shield_hit.wav  
        weapon_upgrade.wav  
        ui_click.wav  
        wave_start.wav  
        game_over.wav  
        combo_up.wav
```

The AudioManager loads clips from Resources at runtime. Missing clips are silently skipped — the game works perfectly without audio files.